



**SILVER OAK
UNIVERSITY**
EDUCATION TO INNOVATION



SILVER OAK UNIVERSITY

Silver Oak College of Computer Application

Course: Foundation of Mathematics | Course Code: MDC5001C | Semester: 1

List of Tutorials

UNIT 2: NUMERICAL METHODS (CO Mapping: CO2)

Tutorial 1: Bracketing Roots — Bisection Method

Topics covered: Solutions of Algebraic and Transcendental Equations — Bisection Method

Q.No	Question	Difficulty	Bloom's Level	CO
1	State the Bisection Method and the condition required for it to be applicable (Intermediate Value Theorem).	Easy	Remember	CO2
2	Explain why the Bisection Method always converges, though slowly.	Easy	Understand	CO2
3	Show that $f(x) = x^3 - x - 1$ has a root between 1 and 2.	Moderate	Apply	CO2
4	Perform 3 iterations of the Bisection Method to find a root of $f(x) = x^2 - 4$ in the interval $[1, 3]$.	Moderate	Apply	CO2
5	Find the root of $f(x) = x^3 - 4x - 9 = 0$ in $[2, 3]$ using 4 iterations of the Bisection Method.	Moderate	Analyze	CO2
6	Estimate the number of iterations required by the Bisection Method to achieve an accuracy of 0.0001 for a root in $[0, 1]$.	Challenging	Analyze	CO2
7	Compare the rate of convergence of the Bisection Method with the Newton-Raphson Method.	Challenging	Evaluate	CO2

Tutorial 2: Refining Roots — Regula-Falsi (False Position) Method

Topics covered: Regula-Falsi Method

Q.No	Question	Difficulty	Bloom's Level	CO
8	State the Regula-Falsi (False Position) formula for finding a root.	Easy	Remember	CO2
9	Explain how the Regula-Falsi method differs from the Bisection method.	Easy	Understand	CO2
10	Find a root of $f(x) = x^3 - 2x - 5 = 0$ in $[2, 3]$ using 2 iterations of Regula-Falsi.	Moderate	Apply	CO2
11	Use the Regula-Falsi method to find the root of $f(x) = \cos(x) - x \cdot e^x$ in $[0, 1]$, up to 3 iterations.	Moderate	Apply	CO2
12	Explain why the Regula-Falsi method can converge more slowly than expected in some cases (one-sided convergence).	Moderate	Analyze	CO2
13	Compare the Bisection and Regula-Falsi methods in terms of convergence speed and reliability.	Challenging	Analyze	CO2
14	Find the root of $x \cdot \log_{10} x - 1.2 = 0$ in $[2, 3]$ using Regula-Falsi, correct to 3 decimal places.	Challenging	Evaluate	CO2

Tutorial 3: Fast Convergence — Newton-Raphson Method

Topics covered: Newton-Raphson Method

Q.No	Question	Difficulty	Bloom's Level	CO
15	State the Newton-Raphson iterative formula.	Easy	Remember	CO2
16	Explain the geometric interpretation of the Newton-Raphson method.	Easy	Understand	CO2
17	Use Newton-Raphson to find $\sqrt{2}$ correct to 4 decimal places, starting with $x_0 = 1.5$.	Moderate	Apply	CO2
18	Find a root of $f(x) = x^3 - 3x - 5 = 0$ near $x_0 = 2$ using 3 iterations of Newton-Raphson.	Moderate	Apply	CO2
19	Discuss the conditions under which the Newton-Raphson method may fail to converge.	Moderate	Analyze	CO2
20	Derive the Newton-Raphson formula using the Taylor series expansion.	Challenging	Analyze	CO2
21	Solve $x \cdot e^x = 1$ using the Newton-Raphson method, correct to 3 decimal places.	Challenging	Evaluate	CO2
22	Compare the order of convergence of the Newton-Raphson method with the Bisection and Regula-Falsi methods.	Challenging	Evaluate	CO2

Tutorial 4: Two-Point Iteration — Secant Method

Topics covered: Secant Method

Q.No	Question	Difficulty	Bloom's Level	CO
23	State the Secant method formula for root finding.	Easy	Remember	CO2
24	Explain how the Secant method avoids computing derivatives, unlike Newton-Raphson.	Easy	Understand	CO2
25	Use the Secant method to find a root of $f(x) = x^3 - x - 1$ with $x_0 = 1$, $x_1 = 2$ (2 iterations).	Moderate	Apply	CO2
26	Find the root of $f(x) = \cos(x) - x$ using the Secant method with initial guesses 0 and 1 (3 iterations).	Moderate	Apply	CO2
27	Compare the Secant method and Regula-Falsi method in terms of formula structure and convergence behavior.	Moderate	Analyze	CO2
28	Explain why the Secant method has a superlinear order of convergence (≈ 1.618), and derive it briefly.	Challenging	Evaluate	CO2

Tutorial 5: Solving Systems — Gauss Elimination Method

Topics covered: Solution of Linear System of Equations — Gauss Elimination

Q.No	Question	Difficulty	Bloom's Level	CO
29	Describe the steps of the Gauss Elimination method for solving a linear system.	Easy	Remember	CO2
30	Define pivoting and explain why it is needed in Gauss Elimination.	Easy	Understand	CO2
31	Solve the system $x+y+z=6$, $2x+3y+z=11$, $x-y+2z=5$ using Gauss Elimination.	Moderate	Apply	CO2
32	Solve using Gauss Elimination: $2x-y+3z=9$, $x+y+z=6$, $x-y+z=2$.	Moderate	Apply	CO2
33	Explain the difference between Gauss Elimination and Gauss-Jordan elimination.	Moderate	Analyze	CO2
34	Explain why partial pivoting is necessary when a pivot element is very small, and illustrate with a 3×3 example.	Challenging	Analyze	CO2
35	Solve using Gauss Elimination: $4x+y-z=13$, $2x-4y+z=-8$, $x+2y+3z=13$.	Challenging	Evaluate	CO2

Tutorial 6: Solving Systems — Gauss-Jordan Method

Topics covered: Gauss-Jordan Method

Q.No	Question	Difficulty	Bloom's Level	CO
36	Describe the Gauss-Jordan elimination method and how it differs from Gauss Elimination.	Easy	Remember	CO2
37	Explain the concept of reduced row echelon form (RREF) in the context of Gauss-Jordan.	Easy	Understand	CO2
38	Solve the system $2x+y-z=3$, $x-y+2z=1$, $3x+2y+z=10$ using Gauss-Jordan.	Moderate	Apply	CO2
39	Use Gauss-Jordan elimination to find the inverse of the matrix $\begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$.	Moderate	Apply	CO2
40	Solve $x+2y+z=3$, $2x+5y-z=-4$, $3x-2y-z=5$ using Gauss-Jordan.	Moderate	Analyze	CO2
41	Discuss the computational cost of Gauss-Jordan compared to Gauss Elimination.	Challenging	Analyze	CO2
42	Explain the general procedure to compute the inverse of a 3×3 matrix using Gauss-Jordan elimination.	Challenging	Evaluate	CO2

Tutorial 7: Iterative Solutions — Gauss-Seidel & Jacobi Methods

Topics covered: Gauss-Seidel and Jacobi Iterative Methods

Q.No	Question	Difficulty	Bloom's Level	CO
43	State the Jacobi iterative formula for solving a system of linear equations.	Easy	Remember	CO2
44	State the Gauss-Seidel iterative formula and explain how it differs from the Jacobi method.	Easy	Understand	CO2
45	Explain the diagonal dominance condition required for convergence of the Jacobi and Gauss-Seidel methods.	Moderate	Understand	CO2
46	Perform 3 iterations of the Jacobi method for $10x+y+z=12$, $x+10y+z=12$, $x+y+10z=12$, starting with $x=y=z=0$.	Moderate	Apply	CO2
47	Perform 3 iterations of the Gauss-Seidel method for the same system as above, starting with $x=y=z=0$.	Moderate	Apply	CO2
48	Compare the rate of convergence of the Jacobi and Gauss-Seidel methods, and explain why Gauss-Seidel typically converges faster.	Challenging	Analyze	CO2

Q.No	Question	Difficulty	Bloom's Level	CO
49	Solve using Gauss-Seidel: $4x+2y+z=14$, $x+5y-z=10$, $x+y+8z=20$, up to 3 iterations.	Challenging	Evaluate	CO2
50	Discuss a scenario where the Jacobi or Gauss-Seidel method may fail to converge, and explain how diagonal dominance addresses this.	Challenging	Evaluate	CO2

Tutorial 8: Estimating Values — Newton's Forward & Backward Interpolation

Topics covered: Newton's Forward and Backward Interpolation

Q.No	Question	Difficulty	Bloom's Level	CO
51	State Newton's Forward Difference Interpolation formula.	Easy	Remember	CO2
52	State Newton's Backward Difference Interpolation formula, and explain when it is preferred over the forward formula.	Easy	Understand	CO2
53	Construct a forward difference table for $x = 1, 2, 3, 4, 5$; $y = 1, 8, 27, 64, 125$.	Moderate	Apply	CO2
54	Using Newton's Forward Interpolation formula, estimate the value of y at $x = 1.5$ for the data in Q59.	Moderate	Apply	CO2
55	Given $x = 10, 20, 30, 40$ with $y = 1.1, 2.0, 4.4, 7.9$, estimate y at $x = 35$ using Newton's Backward Interpolation.	Moderate	Analyze	CO2
56	Explain the difference between interpolation and extrapolation, and state which formula is better suited near the end of a data table.	Challenging	Analyze	CO2
57	Derive Newton's Forward Interpolation formula using the forward difference operator.	Challenging	Evaluate	CO2
58	For a table of 6 equally spaced points, explain which formula (forward/backward) should be used to estimate a value near the start of the table, with justification.	Challenging	Evaluate	CO2

Tutorial 9: Unequal Intervals — Lagrange's Interpolation

Topics covered: Lagrange's Interpolation

Q.No	Question	Difficulty	Bloom's Level	CO
59	State Lagrange's Interpolation formula for n data points.	Easy	Remember	CO2

Q.No	Question	Difficulty	Bloom's Level	CO
60	Explain why Lagrange's Interpolation is used for data with unequally spaced x-values.	Easy	Understand	CO2
61	Given the points (1,3), (3,9), (4,30), find the Lagrange interpolating polynomial and estimate y at x=2.	Moderate	Apply	CO2
62	Using Lagrange's formula, find the interpolating polynomial for the points (0,2), (1,3), (3,5).	Moderate	Apply	CO2
63	Compare Lagrange's Interpolation with Newton's Divided Difference Interpolation in terms of computational efficiency when a new data point is added.	Moderate	Analyze	CO2
64	Estimate f(2.5) given data points (1,1), (2,4), (3,9), (4,16) using Lagrange's Interpolation.	Challenging	Evaluate	CO2
65	Derive the general form of the Lagrange basis polynomials $L_i(x)$, and verify $L_i(x_j) = 1$ if $i=j$ and 0 otherwise.	Challenging	Create	CO2

Tutorial 10: Divided Differences — Newton's Divided Difference Interpolation

Topics covered: Newton's Divided Difference Interpolation

Q.No	Question	Difficulty	Bloom's Level	CO
66	Define the divided difference of a function and state its formula for two points.	Easy	Remember	CO2
67	Construct a divided difference table for the data (2,4), (4,56), (5,105), (6,180).	Easy	Understand	CO2
68	Using Newton's Divided Difference formula, find the interpolating polynomial for the data in Q73 and estimate f(3).	Moderate	Apply	CO2
69	Given points (1,1), (3,9), (5,25), (7,49), construct the divided difference table and find the interpolating polynomial.	Moderate	Apply	CO2
70	Explain the advantage of Newton's Divided Difference method over the Lagrange method when a new data point is added.	Moderate	Analyze	CO2
71	Show that Newton's Divided Difference formula reduces to Newton's Forward Difference formula when the data points are equally spaced.	Challenging	Analyze	CO2
72	Using Newton's Divided Difference Interpolation, estimate f(5) given (1,2), (2,4), (4,16), (7,89).	Challenging	Evaluate	CO2

